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[Intervention Review]

Advanced sperm selection techniques for assisted reproduction

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ABSTRACT

Rationale

Assisted reproductive technologies (ART), including in vitro fertilisation (IVF) and intracytoplasmic sperm injection (ICSI), combine an egg and sperm outside the body to enhance the probability of fertilisation and pregnancy. Advanced sperm selection techniques are increasingly employed in ART, most commonly in cycles utilising ICSI. Advanced sperm selection techniques are hypothesised to improve the likelihood that structurally intact and mature sperm with high DNA integrity are selected. Strategies include selection according to surface charge; sperm apoptosis; sperm birefringence; magnetic-activated cell sorting; ability to bind to hyaluronic acid; microfluidic selection; and sperm morphology under ultra-high magnification.

This is an update of a Cochrane review first published in 2014 and updated in 2019.

Objectives

To evaluate the benefits and harms of advanced sperm selection techniques on assisted reproductive technologies (ART) outcomes.

Search methods

We conducted a systematic search of electronic databases (CENTRAL via the Cochrane Register of Studies Online, MEDLINE, and Embase) for relevant randomised controlled trials (RCTs). We handsearched the reference lists of included studies and similar reviews. The search was conducted on 24 February 2025.

Eligibility criteria

We included randomised controlled trials (RCTs) comparing advanced sperm selection techniques versus standard IVF or ICSI, or versus another advanced sperm selection technique. We excluded studies on intracytoplasmic morphologically selected sperm injection (IMSI), as they are the subject of a separate Cochrane review. Due to increasing concerns about untrustworthy data in health research, we carried out the Trustworthiness in Randomised Controlled Trials (TRACT) assessment for all eligible studies.

Outcomes

Our primary outcomes were live birth and miscarriage per woman randomly assigned. Secondary outcomes included clinical pregnancy per woman randomly assigned, miscarriage per clinical pregnancy, and foetal abnormalities.

Risk of bias

Two review authors independently assessed the risk of bias in included studies using the Cochrane RoB 2 tool. Any disagreements were resolved by a third review author.

Synthesis methods

Two review authors independently assessed study eligibility and extracted data. Any disagreements were resolved by a third review author. We consulted study investigators to answer queries. Risk ratios (RRs) were calculated with 95% confidence intervals (CIs). We combined studies' data using the fixed-effect model. We evaluated the certainty of the evidence using GRADE methods.

Included studies

We found one new study to include in this updated review. We also removed four previously included studies as we had concerns about the trustworthiness of their data. Therefore, in this update, we included five RCTs (3752 women).

Our assessment of the certainty of evidence ranged from very low to moderate. The main limitations in the evidence were performance bias and attrition bias.

The outcome of foetal abnormalities was not reported in the included studies.

Synthesis of results

Hyaluronic acid selected sperm-intracytoplasmic sperm injection (HA-ICSI) versus ICSI

One RCT evaluated the effects of HA-ICSI versus ICSI on live birth. HA-ICSI may result in no difference or a slight increase in live birth per woman randomly assigned: there was a 25% chance of live birth with ICSI versus 24% to 31% with HA-ICSI (RR 1.09, 95% CI 0.97 to 1.24; I^2 not applicable; 1 study, 2772 participants; low-certainty evidence). Two RCTs reported on miscarriage. HA-ICSI probably slightly decreases miscarriage per woman randomly assigned: 7% chance of miscarriage with ICSI versus 3% to 6% chance with HA-ICSI (RR 0.59, 95% CI 0.44 to 0.80; $I^2 = 0\%$; 2 studies, 3327 participants; moderate-certainty evidence) and per clinical pregnancy: 18% chance of miscarriage with ICSI compared to 8% to 15% chance with HA-ICSI (RR 0.60, 95% CI 0.45 to 0.80; $I^2 = 0\%$; 2 studies, 1207 participants; moderate-certainty evidence). Two RCTs reported on clinical pregnancy. There may be little or no difference between groups: 37% chance of pregnancy with ICSI versus 33% to 40% chance with HA-ICSI (RR 0.98, 95% CI 0.90 to 1.07; $I^2 = 0\%$; 2 studies, 3327 participants; low-certainty evidence).

HA-ICSI versus SpermSlow

One RCT compared two advanced sperm selection techniques: HA-ICSI versus SpermSlow. The certainty of the evidence is very low. We are uncertain about the relative effects of these techniques on live birth (RR 1.13, 95% CI 0.64 to 2.01; 1 study, 100 participants), miscarriage (per woman (RR 0.80, 95% CI 0.23 to 2.81; 1 study, 100 participants) and per clinical pregnancy (RR 0.76, 95% CI 0.24 to 2.44; 1 study, 41 participants)), and clinical pregnancy (RR 1.05, 95% CI 0.66 to 1.68; 1 study, 100 participants).

Zeta potential sperm selection versus ICSI

One RCT evaluated zeta potential sperm selection versus ICSI. The certainty of the evidence is very low. We are uncertain of the effect of zeta potential sperm selection on live birth (RR 2.48, 95% CI 1.34 to 4.56; 1 study, 203 participants), miscarriage (per woman (RR 0.74, 95% CI 0.17 to 3.23; 1 study, 203 participants) and per clinical pregnancy (RR 0.41, 95% CI 0.10 to 1.68; 1 study, 62 participants)), and clinical pregnancy (RR 1.80, 95% CI 1.16 to 2.80; 1 study, 203 participants).

Microfluidic Sperm Sorting (MFSS) versus ICSI

One RCT evaluated MFSS versus ICSI. The certainty of the evidence is very low. We are uncertain of the effect of MFSS on live birth (RR 1.10, 95% CI 0.68 to 1.76; 1 study, 122 participants), miscarriage (per woman (RR 1.20, 95% CI 0.39 to 3.72; 1 study, 122 participants) and per clinical pregnancy (RR 1.08, 95% CI 0.37 to 3.11; 1 study, 55 participants)), and clinical pregnancy (RR 1.12, 95% CI 0.75 to 1.65; 1 study, 122 participants).

Authors' conclusions

The evidence suggests that use of HA-ICSI may make no difference to or may slightly increase live-birth rate and probably slightly reduces miscarriage rate. We do not know the effect of the other advanced sperm selection techniques on live birth or miscarriage, or clinical pregnancy, principally because of the very low certainty of the evidence.

Further high-quality studies, including awaited data from 19 ongoing studies, are required to evaluate whether any of these advanced sperm selection techniques can be recommended for use in routine practice.

Funding

This Cochrane review had no dedicated funding.

Registration

Original review (2014) DOI: 10.1002/14651858.CD010461.pub2

PLAIN LANGUAGE SUMMARY

Does the use of advanced sperm selection techniques improve the effectiveness of assisted reproduction for women trying to have a baby?

Key messages

- Advanced sperm selection techniques can be used in patients trying to get pregnant using assisted reproduction procedures (vitro fertilisation (IVF) and intracytoplasmic sperm injection (ICSI)). The aim of using the advanced techniques is to increase the chances of selecting the best sperm, leading to successful fertilisation, but the current evidence is unclear.
- Selecting sperm that attach to a natural substance in the body called hyaluronic acid (HA-ICSI (hyaluronic acid selected sperm-intracytoplasmic sperm injection)) probably reduces the risk of miscarriage slightly compared to the standard method of sperm selection. HA-ICSI may make no difference to or may slightly increase the likelihood of delivering a living baby.
- Evidence for other techniques is very limited. More studies are needed to find out whether advanced sperm selection techniques make IVF more effective than the standard method currently used.

What is assisted reproductive technology?

Assisted reproductive technology is used to bring about pregnancy by artificial means. In vitro fertilisation (IVF), collection of a woman's egg or eggs from the ovaries then fertilisation with sperm in a laboratory, is a commonly used treatment for people diagnosed with infertility who want to get pregnant. Intracytoplasmic sperm injection (ICSI) is sometimes used as well. In ICSI, a single sperm is directly injected into the egg. Currently, IVF laboratories use standard methods to select healthy-looking sperm for fertilisation. It is thought that selecting sperm of high quality may improve the chances of pregnancy and the birth of a healthy baby.

What are advanced sperm selection techniques?

Advanced sperm selection techniques are complex methods used to select sperm for fertilisation that are healthy, mature, and structurally sound (the right shape and size). These techniques can be used to improve sperm selection in infertility that results from male factors (such as low sperm count or sluggish sperm). Despite the use of these techniques in many assisted reproduction centres worldwide, their effectiveness is unclear.

What did we want to find out?

We wanted to find out which, if any, of the advanced sperm selection techniques that are used for assisted reproduction increase the chances of:

- live birth (giving birth to a living baby);
- miscarriage (losing the baby before 20 weeks of development in the womb);
- clinical pregnancy (having a positive pregnancy test confirmed by ultrasound);
- foetal abnormalities (something wrong with the baby).

This review is the second update of a review originally published in 2014 and first updated in 2019.

What did we do?

We searched medical databases online for randomised studies that compared advanced sperm selection techniques in IVF to standard methods or to other advanced sperm selection techniques. In randomised studies, participants are randomly allocated to two or more groups to ensure that their treatment is similar other than the aspect that is being compared (sperm selection in this case). We then combined results from all relevant studies and rated our confidence in the evidence based on the quality of the studies (e.g. the way they were designed and the number of people they involved).

What did we find?

We found five studies that involved a total of 3752 women. Two studies evaluated sperm selection by the sperm's ability to bind a substance called hyaluronic acid during the process of ICSI (HA-ICSI) versus ICSI on its own. One study compared HA-ICSI to a technique called SpermSlow (a special solution that helps to select healthier sperm). One study evaluated selecting sperm that have a slight electrical

charge on their surface ('zeta potential') versus ICSI. One study evaluated sperm selected by a sorting system called Microfluidic Sperm Sorting versus ICSI.

Four of the studies reported live birth; four reported miscarriage; five reported clinical pregnancy; and none reported whether there were foetal abnormalities.

Main results

The studies in our review showed that, in comparison to ICSI, HA-ICSI:

- may result in no difference or a slight increase in live births (1 study, 2772 people);
- probably slightly decreases miscarriage (2 studies, 3327 people); and
- may make no difference to the chance of getting pregnant (2 studies, 3327 women).

We are uncertain whether the other sperm selection techniques affect live birth, miscarriage, or pregnancy rates.

No studies reported on foetal abnormalities.

Further studies of suitable quality are required before any of these advanced sperm selection techniques can be recommended for use in clinical practice.

What are the limitations of the evidence?

The main limitations were due to the small numbers of people taking part in the studies. It is possible that people in the studies were aware of the treatment being given, which may have had an effect on how the results were reported. Not all studies covered important things to consider when people undergo assisted reproduction. For example, there was no information about foetal abnormalities. More high-quality studies with a large number of participants are needed to know whether advanced sperm selection techniques are worthwhile for patients undergoing IVF.

How up to date is this evidence?

This review updates our previous review. The evidence is current to February 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Hyaluronic acid-selected sperm (HA-ICSI) compared to ICSI for assisted reproduction

Hyaluronic acid-selected sperm (HA-ICSI) compared to ICSI for assisted reproduction

Patient or population: assisted reproduction

Setting: Assisted Reproduction Clinic

Intervention: Hyaluronic acid-selected sperm (HA-ICSI)

Comparison: ICSI

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with ICSI	Risk with Hyaluronic acid-selected sperm (HA-ICSI)				
Live birth per woman randomly assigned	25 per 100	27 per 100 (24 to 31)	RR 1.09 (0.97 to 1.24)	2772 (1 RCT)	⊕⊕⊕⊖ Low ^{a,b}	HA-ICSI may result in no difference or a slight increase in live birth per woman randomly assigned.
Miscarriage per woman randomly assigned	7 per 100	4 per 100 (3 to 5)	RR 0.59 (0.44 to 0.80)	3327 (2 RCTs)	⊕⊕⊕⊖ Moderate ^a	HA-ICSI probably slightly reduces miscarriage per woman randomly assigned.
Miscarriage per clinical pregnancy	18 per 100	11 per 100 (8 to 14)	RR 0.60 (0.45 to 0.80)	1207 (2 RCTs)	⊕⊕⊕⊖ Moderate ^a	HA-ICSI probably slightly reduces miscarriage per clinical pregnancy.
Clinical pregnancy per woman randomly assigned	37 per 100	36 per 100 (33 to 39)	RR 0.98 (0.90 to 1.07)	3327 (2 RCTs)	⊕⊕⊕⊖ Low ^{a,b}	The evidence suggests that hyaluronic acid-selected sperm (HA-ICSI) may result in little to no difference in clinical pregnancy per woman randomly assigned.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_461187969640585253.

^a Downgraded once for risk of bias. Both studies have domains that are at high risk of bias.

^b Downgraded once due to imprecision. The effect estimate crossing the null and including no effect

Summary of findings 2. Summary of findings table - Hyaluronic acid-selected sperm (HA-ICSI) compared to SpermSlow for assisted reproduction

Hyaluronic acid-selected sperm (HA-ICSI) compared to SpermSlow for assisted reproduction

Patient or population: assisted reproduction

Setting: Assisted Reproduction Clinic

Intervention: Hyaluronic acid-selected sperm (HA-ICSI)

Comparison: SpermSlow

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with SpermSlow	Risk with Hyaluronic acid-selected sperm (HA-ICSI)				
Live birth per woman randomly assigned	30 per 100	34 per 100 (19 to 60)	RR 1.13 (0.64 to 2.01)	100 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of HA-ICSI on live birth per woman randomly assigned.
Miscarriage per woman randomly assigned	10 per 100	8 per 100 (2 to 28)	RR 0.80 (0.23 to 2.81)	100 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of HA-ICSI on miscarriage per woman randomly assigned.
Miscarriage per clinical pregnancy	25 per 100	19 per 100 (6 to 61)	RR 0.76 (0.24 to 2.44)	41 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of HA-ICSI on miscarriage per clinical pregnancy.
Clinical pregnancy per woman randomly assigned	40 per 100	42 per 100 (26 to 67)	RR 1.05 (0.66 to 1.68)	100 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of HA-ICSI on clinical pregnancy per woman randomly assigned.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

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See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_461188136309905517.

^a Downgraded twice for high risk of bias. Single study only and it has more than one domain rated as high risk or unclear risk.

^b Downgraded twice for imprecision. Small sample size, number of events are low and CI are very wide. Point estimate crossed the line of no effect.

Summary of findings 3. Summary of findings table - Zeta potential sperm selection compared to ICSI for assisted reproduction

Zeta potential sperm selection compared to ICSI for assisted reproduction

Patient or population: assisted reproduction

Setting: Assisted Reproduction Clinic

Intervention: zeta potential sperm selection

Comparison: ICSI

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with ICSI	Risk with zeta potential sperm selection				
Live birth per woman randomly assigned	12 per 100	29 per 100 (16 to 54)	RR 2.48 (1.34 to 4.56)	203 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of zeta potential sperm selection on live birth per woman randomly assigned.
Miscarriage per woman randomly assigned	4 per 100	3 per 100 (1 to 13)	RR 0.74 (0.17 to 3.23)	203 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,c}	The evidence is very uncertain about the effect of zeta potential sperm selection on miscarriage per woman randomly assigned.
Miscarriage per clinical pregnancy	18 per 100	7 per 100 (2 to 31)	RR 0.41 (0.10 to 1.68)	62 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,c}	The evidence is very uncertain about the effect of zeta potential sperm selection on miscarriage per clinical pregnancy.
Clinical pregnancy per woman randomly assigned	22 per 100	39 per 100 (25 to 61)	RR 1.80 (1.16 to 2.80)	203 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of zeta potential sperm selection on clinical pregnancy per woman randomly assigned.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

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See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_461188177150105752.

^a Downgraded once for risk of bias. Only one study was included for this outcome, and there is an unclear risk of selection bias.

^b Downgraded twice for imprecision. Only one study was included for this outcome. The sample size and number of events are low.

^c Downgraded twice for imprecision. Only one study was included for this outcome. The sample size and number of events are low, and the confidence interval includes both harms and benefits.

Summary of findings 4. Summary of findings table - Microfluidic Sperm Sorting (MFSS) compared to ICSI for assisted reproduction

Microfluidic Sperm Sorting (MFSS) compared to ICSI for assisted reproduction

Patient or population: assisted reproduction

Setting: Assisted Reproduction Clinic

Intervention: Microfluidic Sperm Sorting (MFSS)

Comparison: ICSI

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with ICSI	Risk with Microfluidic Sperm Sorting (MFSS)				
Live birth per woman randomly assigned	34 per 100	38 per 100 (23 to 61)	RR 1.10 (0.68 to 1.76)	122 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of MFSS on live birth per woman randomly assigned.
Miscarriage per woman randomly assigned	8 per 100	10 per 100 (3 to 30)	RR 1.20 (0.39 to 3.72)	122 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of MFSS on miscarriage per woman randomly assigned.

Miscarriage per clinical pregnancy	19 per 100	21 per 100 (7 to 60)	RR 1.08 (0.37 to 3.11)	55 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of MFSS on miscarriage per clinical pregnancy.
Clinical pregnancy per woman randomly assigned	43 per 100	48 per 100 (32 to 70)	RR 1.12 (0.75 to 1.65)	122 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence is very uncertain about the effect of MFSS on clinical pregnancy per woman randomly assigned.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

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See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_461187996450838580.

^a Downgraded once for risk of bias. There is only one study for this outcomes and there is an unclear risk of bias for selection and performance bias.

^b Downgraded twice for imprecision. Only one study included for this outcome. The sample size and event rate is low and the point estimate include both benefits and harms.

BACKGROUND

Description of the condition

In vitro fertilisation (IVF) is a form of assisted reproductive technology (ART) that is used for treating infertility, a condition that affects an estimated 8% to 12% of the population [1]. IVF usually involves controlled ovarian stimulation (COS), surgical oocyte retrieval, in vitro fertilisation, and embryo transfer. Intracytoplasmic sperm injection (ICSI) involves injecting a single sperm into the cytoplasm of each oocyte to achieve fertilisation [2]. Intracytoplasmic sperm injection is commonly used as a treatment for male factor infertility when semen parameters are poor, when sperm have been surgically retrieved, or when repeated fertilisation with standard IVF has failed [3].

Despite technological advances, pregnancy and live birth rates from ART remain relatively low, hence the drive for researchers to identify other modifiable aetiologies such as sperm dysfunction. Successful embryo development and subsequent pregnancy outcome are likely to be impacted by the quality of the sperm that fertilise an oocyte [4]. Ideally, only sperm with a high chance of successful fertilisation and subsequent embryo growth would be used for ART. These sperm would be viable and mature, have high DNA integrity, and be structurally sound [5].

Sperm preparation and selection in IVF are limited to semen washing, density gradient centrifugation, and use of swim-up techniques [6]. In ICSI, routine sperm selection is based on motility and gross morphology (sperm are examined under a microscope at 200× to 400× magnification) after one or more of the above methods of semen preparation has been applied [5]. Advanced sperm selection techniques based on alternative characteristics might enable further selection of the most appropriate sperm for use in ART, but there are limited data on the efficacy of other sperm selection techniques. There is no guidance from major clinical organisations on the use of advanced sperm selection techniques in routine practice.

Description of the intervention and how it might work

Advanced sperm selection techniques have been developed with the aim of improving ART outcomes in certain clinical scenarios. The techniques can be categorised as follows.

Surface charge selection

Electrophoretic sperm selection and sperm zeta potential are surface charge selection protocols utilised in both IVF and ICSI. The zeta potential of the sperm is the electrical potential between the sperm membrane and its surroundings. The zeta potential decreases with capacitation, and normally differentiated sperm are charged electronegatively. Semen is placed into an electrophoretic device and a current is applied. Normally differentiated negatively charged sperm are rapidly separated and collected from an adjacent chamber [7].

Sperm apoptosis

Selection of non-apoptotic sperm for use in ART is based on the presence of phosphatidylserine on the external surface of the sperm membrane in the early stages of apoptosis. Magnetic-activated cell sorting (MACS) and glass wool separation columns utilise the magnetic properties of phosphatidylserine to separate apoptotic sperm from non-apoptotic sperm [8].

Hyaluronic acid binding

Hyaluronic acid (HA) is the main component of the extracellular matrix of the cumulus oophorus. Hyaluronic acid binding sites on the sperm plasma membrane indicate sperm maturity. Mature sperm bind to and digest HA and thus have a better chance of reaching the oocyte for fertilisation. In vitro, HA is utilised as a 'physiological selector' of mature intact sperm [5, 9].

Two systems for HA sperm selection are currently available. Physiological intracytoplasmic sperm injection (PICSI; Origio, Måløv, Denmark) is a plastic culture dish with spots of HA attached to its base. Sperm are bound by the head to HA and are selected for microinjection [10]. SpermSlow is a viscous medium containing HA. Appropriate sperm appear 'slowed' and are selected [11].

Sperm birefringence

The mature sperm nucleus has high intrinsic birefringence due to longitudinally orientated subacrosomal protein filaments. With the use of polarised light microscopy, sperm birefringence can be evaluated and a mature sperm selected [12].

Microfluidic Sperm Sorting (MFSS)

This technique uses a chip to select sperm with a view to mimicking the in vivo female reproductive tract and aims to control the behaviour of sub-microlitre volumes of fluid. Minute volumes of liquids run in small-scale microchannel networks and allow selection of sperm based on motility and behavioural properties. The goal is to select motile sperm with high DNA integrity. Of note, there is no standardised chip, and current MFSS technologies are all slightly different [13].

Sperm morphology (intracytoplasmic morphologically selected sperm injection)

Subtle defects in sperm morphology (acrosome, nucleus, mitochondria, tail, postacrosomal lamina, and neck) can be observed using ultra-high magnification (6000×) microscopy (motile sperm organelle morphology examination (MSOME)) [14]. Intracytoplasmic morphologically selected sperm injection (IMSI) is a modification of ICSI that uses this technique [15]. This review did not evaluate IMSI, as it is the subject of another Cochrane review. The evidence from randomised controlled trials did not support or refute the clinical use of IMSI [16].

Each sperm selection modality utilises different characteristics of sperm structure, physiology, or function to promote selection of the most normal sperm. Selection of the most appropriate sperm for fertilisation in vitro may help improve fertilisation and the quality of embryos created, therefore providing a better chance of a healthy pregnancy. Advanced sperm selection protocols aim to improve ART outcomes and may limit the possible deleterious effects on offspring of using sperm with defective DNA [17].

Why it is important to do this review

Advanced sperm selection techniques are hypothesised to improve ART outcome through the selection of sperm with a variety of 'beneficial characteristics'. Although individual small studies have suggested that these techniques have clinical benefit [18], there remains no comprehensive review of randomised controlled trials (RCTs) in this area. This is an update of a Cochrane review first published in 2014 [19], and previously updated in 2019 [20]. Since

then, new RCTs have been conducted exploring advanced sperm selection and new techniques such as MFSS have been developed, warranting an updated review on this topic.

OBJECTIVES

To evaluate the benefits and harms of advanced sperm selection techniques on assisted reproductive technology outcomes.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting [21, 22].

Differences between protocol and review or between versions

Following completion of our protocol [23], a Cochrane review titled 'Regular (ICSI) versus ultra-high magnification (IMSI) sperm selection for assisted reproduction' was published [16]. This publication overlapped considerably with our protocol, and after consultation with the Cochrane Gynaecology and Fertility Group, we amended the scope of our review to exclude the use of IMSI for sperm selection. The title of our review was amended accordingly [19].

In the 2019 update of our review, we added the adverse outcome of miscarriage per woman randomly assigned as a primary outcome [20].

In the 2026 update of our review, we added a trustworthiness assessment to the protocol. Due to increasing concerns about research misconduct, such as data fabrication in the scientific community, we applied the Trustworthiness in RAndomised Controlled Trials (TRACT) checklist [24] to all included studies from both the 2019 update and this new update.

Any studies that commenced after January 2010 and did not have a trial registration record, or had potential trustworthiness concerns (as flagged by the TRACT checklist) were categorised as 'Awaiting classification' unless the trial authors could provide one of the following to confirm trustworthiness and timeliness of the trial:

- official award letter from a study funder;
- date-stamped ethics approval notification for the trial;
- official email from an Independent Data Monitoring Committee to the trial team; or
- the individual participant data for the study (i.e. a spreadsheet with all the rows of data).

A six-month buffer period was permitted; trials registered within six months of commencing recruitment were exempt from the above requirement.

Any trustworthiness concerns flagged by the TRACT checklist can be found in [Supplementary material 2](#) under 'Additional comments and concerns'.

Criteria for considering studies for this review

Types of studies

Published and unpublished randomised controlled trials (RCTs) investigating the impact of advanced sperm selection techniques in assisted reproduction technology (ART) were eligible for inclusion.

Non-randomised studies and cross-over studies were not eligible for inclusion.

Types of participants

Participants were women undergoing assisted reproductive technologies.

Types of interventions

Trials comparing an advanced sperm selection technique either with another advanced sperm selection technique or with standard sperm preparation techniques (e.g. semen washing, density gradient centrifugation, swim-up techniques).

Advanced sperm selection techniques include the following.

- Surface charge selection
- Sperm apoptosis
- Hyaluronic acid (HA) binding (HA-ICSI)
- Magnetic activated cell sorting (MACS)
- Sperm birefringence
- Microfluidic Sperm Sorting (MFSS)

We excluded sperm selection by sperm morphology using ultra-high magnification (IMSI), as this is the subject of another Cochrane review [16].

Outcome measures

We excluded trials that reported none of our outcomes or inadequately reported outcomes.

Critical outcomes

Effectiveness

- Live birth per woman randomly assigned

Live birth is defined as the delivery of a live foetus beyond 20 completed weeks' gestation.

Adverse events

- Miscarriage per woman randomly assigned

Miscarriage is defined as pregnancy loss at less than 20 completed weeks' gestation, or when the foetus weighs less than 500 g. Miscarriage must be confirmed by ultrasound and pregnancy test or histology and includes partial loss of multiple pregnancies.

Important outcomes

Effectiveness

- Clinical pregnancy per woman randomly assigned

Clinical pregnancy is defined as identification of a gestational sac on ultrasound at equal to or greater than seven weeks' gestation.

Adverse events

- Foetal abnormalities per woman randomly assigned

Foetal abnormalities are defined as significant structural, functional, or genetic anomalies that occur during embryogenesis and ongoing gestational development and are identified either during pregnancy, at birth, or later.

- Miscarriage per clinical pregnancy
- Foetal abnormalities per clinical pregnancy

Other outcomes

Fertilisation rates, implantation rates, and outcomes related to embryo development and quality are also of importance in this review and are described in the [Supplementary material 2](#) section. These outcomes were not included in the meta-analyses because standardised grading systems for morphology are lacking, and denominators for fertilisation and implantation rates differ.

Search methods for identification of studies

The search had no language restriction and was designed and conducted by AG and SL, in consultation with the Information Specialist from Cochrane Gynaecology and Fertility. The search was conducted on 24 February 2025.

Electronic searches

We searched the following electronic databases, trial registers, and websites.

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 2) via the Cochrane Register of Studies Online (CRSO); web

platform, searched 24 February 2025 ([Supplementary material 1](#))

- MEDLINE; Ovid platform, searched from 1 January 2018 to 24 February 25 ([Supplementary material 1](#))
- Embase; Ovid platform, searched from 1 January 2018 to 24 February 25 ([Supplementary material 1](#))

ClinicalTrials.gov and the World Health Organisation International Trials Registry Platform search portal are now incorporated into CENTRAL.

Searching other resources

We searched the reference lists of articles retrieved by the search.

Data collection and analysis

For this 2026 update, we used the methods described below to assess suitable studies.

Selection of studies

After an initial screen of titles and abstracts retrieved by the search, we retrieved the full texts of all potentially eligible studies. At least two review authors (from AG, SL, ON, MS, and AA) independently examined these full-text articles for compliance with the inclusion criteria and selected studies eligible for inclusion in the review. We corresponded with study investigators as required to clarify study eligibility. Any disagreements as to study eligibility were resolved by discussion or by consultation with a third review author (SL). We documented the selection process using a PRISMA flow chart ([Figure 1](#)).

Figure 1. PRISMA study flow diagram

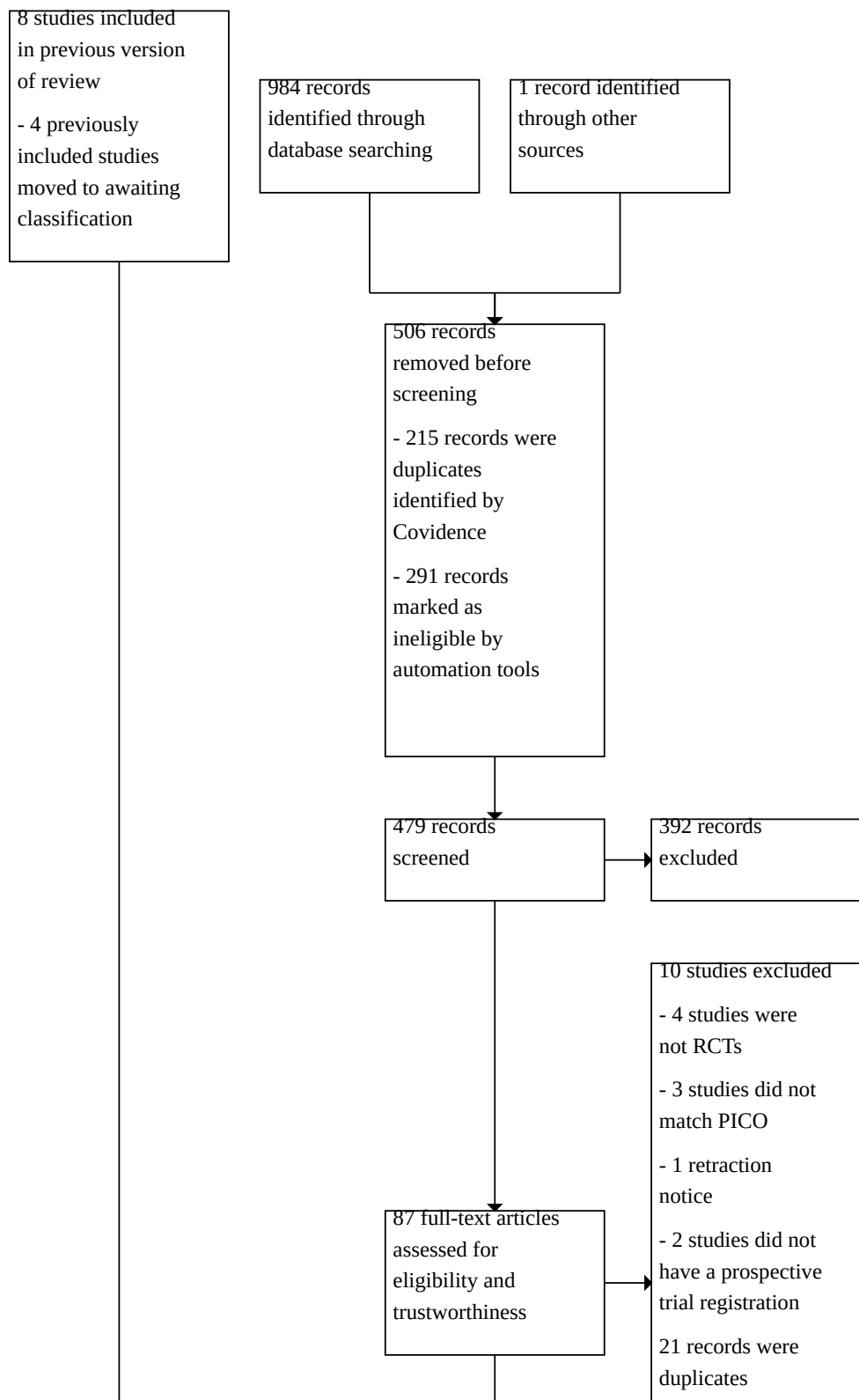
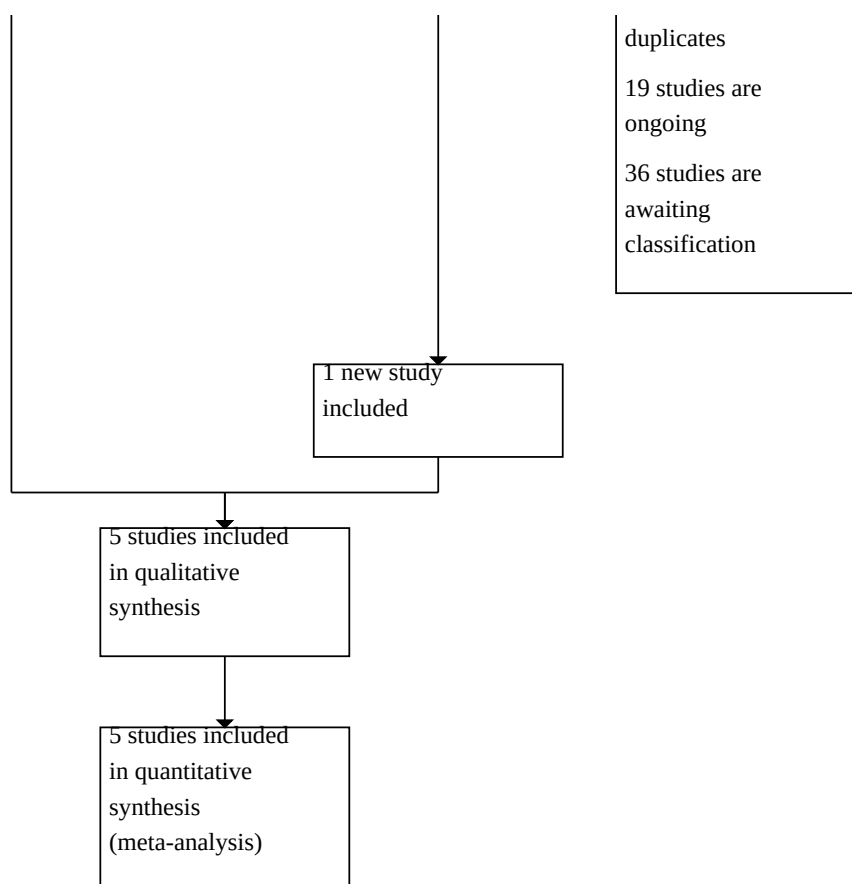


Figure 1. (Continued)



Data extraction and management

Two review authors independently extracted data from the eligible studies (from MS, ON, and AA). Any disagreements were resolved by discussion or by consultation with a third review author (SL). Data extracted included study characteristics and outcome data and details of methods, participants, setting, context, interventions (sperm selection protocols), outcomes, results, and publications. We attempted to contact study investigators via email to obtain additional information. No replies were received from any of the study authors.

Risk of bias assessment in included studies

Two review authors (from MS, ON, and AA) independently assessed the included studies for risk of bias using Cochrane's risk of bias assessment tool RoB 1 [25, 26]. The domains of this instrument are random sequence generation and allocation concealment (selection bias); blinding of participants and personnel; blinding of outcome assessors; incomplete outcome data; selective reporting; and other bias. Any disagreements were resolved by discussion or by consultation with a third review author (SL). We provided support for our judgements and presented our conclusions in the 'risk of bias' table, which we incorporated into the interpretation of review findings by means of sensitivity analysis.

We took care to search for within-trial selective reporting, such as trials failing to report obvious outcomes or reporting outcomes in insufficient detail to allow inclusion in the review. We sought published protocols and compared outcomes between the protocols and the final published studies.

Measures of treatment effect

The extracted data were dichotomous (e.g. live-birth rate, miscarriage rate). Using Review Manager (RevMan) software [27], we entered the numbers of events in the control and intervention groups of each study to calculate Mantel-Haenszel risk ratios (RRs). We presented 95% confidence intervals (CIs) for all outcomes.

Unit of analysis issues

The unit of analysis for our primary analysis was the randomly assigned study participant. Per-pregnancy data were included for some miscarriage outcomes. We briefly summarised data that did not allow valid analysis (e.g. 'per-cycle' data) but did not meta-analyse these data. If studies reported only per-cycle data, we attempted to contact the study authors to obtain 'per-patient randomised' data.

We counted multiple live births (e.g. twins, triplets) as a single live-birth event.

Dealing with missing data

We analysed the data on an intention-to-treat basis to the greatest degree possible and made attempts to obtain missing data from the original trial authors. When the data could not be obtained, we assumed that the outcome (e.g. live birth, clinical pregnancy) did not occur. For other outcomes, we analysed available data. We had planned to subject any imputation undertaken to sensitivity analysis (see [Sensitivity analysis](#)).

Reporting bias assessment

We minimised reporting bias by ensuring that we conducted a comprehensive search for eligible studies. We had planned that if 10 or more studies were included in an analysis, we would use a funnel plot to explore the possibility of small-study effects (a tendency for estimates of the intervention effect to be more beneficial in smaller studies).

Synthesis methods

Where studies were sufficiently similar, we planned to combine data using the fixed-effect model for the review comparisons, that is:

- Advanced sperm selection technique, stratified by individual sperm selection technique versus standard sperm preparation (ICSI) (refer to [Description of the intervention and how it might work](#) for details)
- One advanced sperm selection technique versus another

We displayed an increase in the risk of a particular outcome graphically (the outcome could be beneficial (e.g. live birth) or detrimental (e.g. adverse effects)) on the forest plot of the meta-analysis to the right of the centre line and a decrease in the risk of a particular outcome to the left of the centre line.

We had planned to calculate number needed to treat for an additional beneficial outcome (NNTB) if we identified significant findings.

We considered whether the clinical and methodological characteristics of the included studies were sufficiently similar for meta-analysis to provide a clinically meaningful summary. We assessed statistical heterogeneity using the I^2 statistic, with an I^2 measurement greater than 50% indicating moderate heterogeneity and an I^2 greater than 60% representing substantial heterogeneity [28, 29]. If we found substantial heterogeneity, we planned to explore possible explanations for it using sensitivity analysis (see [Sensitivity analysis](#)) and to consider subgroup analysis. We planned to take any statistical heterogeneity into account when interpreting the results, especially if any variation in the direction of effect was noted.

Investigation of heterogeneity and subgroup analysis

We planned to conduct subgroup analyses to identify separate evidence within the following subgroups, if sufficient data were available.

- Sperm morphology: when the Kruger score is equal to or less than 4%.
- Increased DNA fragmentation index (according to the study cut-off)
- Surgically retrieved sperm

- Female participants over 38 years of age

We did not ultimately perform subgroup analysis once data were extracted and reviewed.

Equity-related assessment

We did not investigate health inequity in this review as it was not part of our protocol.

Sensitivity analysis

We planned to conduct sensitivity analyses for primary outcome measures to determine whether the conclusions were robust to arbitrary decisions made regarding eligibility and analysis. These analyses would have allowed us to consider whether the review's conclusions would have differed if:

- eligibility had been restricted to studies without high risk of bias;
- the random-effects model had been adopted for meta-analysis; or
- alternative imputation strategies had been implemented.

We did not conduct any sensitivity analysis.

Certainty of the evidence assessment

We generated a summary of findings (SoF) table using GRADEproGDT software [30]. In this table, we evaluated the overall certainty of the body of evidence for the main review outcomes (live birth, miscarriage, clinical pregnancy) using GRADE criteria (study limitations (i.e. risk of bias), consistency of effect, imprecision, indirectness, and publication bias). We justified, documented, and incorporated our judgements about the certainty of evidence (high, moderate, low, or very low) into the reporting of results for each outcome.

Consumer involvement

We did not involve consumers in this review due to limited resources, although we did use core outcome sets for the review's outcomes, which were developed with consumer input.

RESULTS

Description of studies

The key features of the included studies are summarised in [Table 1](#).

Results of the search

The search strategy for the initial review identified 1007 studies. Thirty studies were potentially eligible and were retrieved in full text. Following publication of our protocol, a Cochrane review was published regarding sperm selection by sperm morphology under ultra-high magnification [16]. After discussion with the Cochrane Gynaecology and Fertility Group, we amended the scope of our review to exclude the use of ultra-high magnification for sperm selection. Two studies met the inclusion criteria for the original review (Parmegiani 2012a [31, 32, 33]; Worriolow 2013 [34, 35]), both of which evaluated hyaluronic acid binding for ICSI. We excluded 17 studies.

The 2019 update included a further 416 abstracts from a search date limited from 1 January 2014 until 14 June 2018; one study initially classified as ongoing reached completion in January 2019 and was

thus included [20]. After duplicates were removed, 332 remained and 21 full-text articles were assessed. We included eight studies in the 2019 update.

We identified 984 records from database and trial register searches in 2025 as outlined in [Electronic searches](#) and one additional study through the manual searching of included studies' reference lists. We removed 506 records (215 were duplicates, and 291 were ineligible). We screened 479 abstracts or trial register records, of which 392 were excluded. We checked 87 full-text articles: 21 records were duplicates, 19 studies were ongoing, and 36 studies await classification. We excluded 10 studies (see [Excluded studies](#)).

We included one new study (Yetkinel 2019 [36, 37]), making a total of five studies in this 2026 update (Esfahani 2016 [38]; Miller 2019 [39, 40, 41, 42]; Parmegiani 2012a; Worrilow 2013; Yetkinel 2019). There were four studies in the 2019 update that we did not bring forward into this version because we considered them untrustworthy after our TRACT assessment (Majumdar 2013 [43]; Romany 2014 [44, 45, 46]; Troya 2015 [47]; Ziarati 2018 [48, 49]) (see PRISMA flow chart in [Figure 1](#)). In this 2026 update, there were no appropriate included studies reporting outcomes assessing magnetic activated cell sorting (MACS). Although this intervention was reported in the prior review, we deemed the previously included studies to be untrustworthy. We identified no suitable studies regarding sperm selection by birefringence. See [Supplementary material 2](#) and [Supplementary material 3](#).

Included studies

We included five RCTs in the review (Esfahani 2016; Miller 2019; Parmegiani 2012a; Worrilow 2013; Yetkinel 2019).

Design

We attempted to contact authors of three included studies to clarify further details of methodology, but no response was received (Parmegiani 2012a; Worrilow 2013; Yetkinel 2019).

Setting

Three of the studies were single-centre studies, which were conducted in Turkey (Yetkinel 2019), Italy (Parmegiani 2012a), and Iran (Esfahani 2016). The other two studies were multicentre studies: one was conducted at 10 IVF units in the USA (Worrilow 2013) and one at 16 IVF units in the UK (Miller 2019).

Participants

- Three studies included participants being treated with HA-ICSI (total of 1712 women) (Miller 2019; Parmegiani 2012a; Worrilow 2013). Inclusion criteria differed for each study but Parmegiani 2012a included people aged < 41 years, sperm number > 1 million and sperm motility > 5%. Worrilow 2013 excluded couples over 40 years of age and if testicular sperm was used, or if HA (hyaluronic acid) binding was < 2%. Miller 2019 included people aged 18 to 43 years with a body mass index (BMI) of 19 to 35 kg/m², follicle-stimulating hormone (FSH) of 3 to 20 mIU/mL or anti-Müllerian hormone (AMH) at least 1.5 pmol/L. Men were included if they had not had a vasovasostomy or been treated for cancer and had been abstinent for at least three days.
- In the study of patients being treated with zeta potential sperm selection (Esfahani 2016), couples were included if the female partner was below 40 years of age with an adequate follicle count, good oocyte quality, and endometrial thickness below 8

mm, and the male partner had at least one semen parameter below WHO 2010 criteria [50].

- Sixty-one couples were treated with Microfluidic Sperm Sorting (MFSS), and 61 were treated with swim-up in one study (Yetkinel 2019). Women were between 30 to 37 years of age, with male partners between 20 to 40 years, and were undergoing their first or second IVF cycle. Patients were included if they had a diagnosis of unexplained fertility (normal assessment of tubal patency, uterine cavity, ovulation function, absence of poor ovarian reserve per the Bologna criteria, normal semen analysis per WHO 2010 criteria [50].

Interventions

- Two studies compared HA-ICSI versus standard ICSI (Miller 2019; Worrilow 2013).
- One study compared HA-ICSI versus SpermSlow (Parmegiani 2012a).
- One study compared zeta potential sperm selection versus standard ICSI (Esfahani 2016).
- One study compared MFSS versus standard ICSI (Yetkinel 2019).

Outcomes

- Four studies reported live birth (Esfahani 2016; Miller 2019; Parmegiani 2012a; Yetkinel 2019).
- Five studies reported miscarriage rate (Esfahani 2016; Miller 2019; Parmegiani 2012a; Worrilow 2013; Yetkinel 2019).
- Five studies reported clinical pregnancy rate (Esfahani 2016; Miller 2019; Parmegiani 2012a; Worrilow 2013; Yetkinel 2019).
- No studies reported on foetal abnormalities.

Studies awaiting classification

For this update, we took a stricter approach to inclusion of studies than previously. We required that studies commencing recruitment from January 2010 onwards had to have been prospectively registered.

All study authors that did not cite a prospective trial registration were contacted for additional information. Most authors were unresponsive or unable to provide the necessary documentation to confirm the trustworthiness of the trials. This resulted in the following.

- We moved four studies that were previously included in the review to 'awaiting classification'. One of these studies evaluated HA-ICSI versus ICSI (Majumdar 2013), and three of them evaluated MACS versus ICSI (Romany 2014; Troya 2015; Ziarati 2018), one of which also evaluated HA-ICSI versus MACS (Troya 2015).
- We placed 32 studies identified in this update in 'awaiting classification'.

Therefore, a total of 36 studies are awaiting classification ([Supplementary material 4](#)).

Ongoing studies

Nineteen studies appear to be ongoing according to either the trial registration record or author correspondence or both ([Supplementary material 5](#)).

Excluded studies

We brought forward 24 excluded studies from the previous version of the review [20], and we excluded an additional 10 studies in this update, making a total of 34 studies excluded for the following reasons (see [Supplementary material 3](#)).

- On full-text review, 12 studies were found not to be RCTs (Berkovitz 2006 [51]; Casciani 2014 [52]; Charehjooy 2014 [53]; Fleming 2007 [54]; Ghosh 2007 [55]; Han 2009 [56]; Norozi-Hafshejani 2022 [57]; Parmegiani 2010c [58]; Salehi 2023 [59]; San Carchenilla 2013 [60]; Stimpfel 2017 [61]; West 2022 [62]).
- Three studies were quasi-randomised (Gianaroli 2008 [12]; Gianaroli 2010 [63]; Jin 2015 [64]).
- One study was retracted (Battista 2017 [65]).
- Two studies lacked prospective registration (Baldini 2020 [66, 67]; Hasanen 2023 [68, 69]).

- One study analysed participants per treatment randomly assigned (Parmegiani 2010b [70]), and despite attempts to contact the study investigators, we were unable to obtain 'per-woman' data.
- Fifteen studies did not match our PICO (participants, interventions, comparisons, outcomes) inclusion criteria (Antinori 2008 [71, 72]; Balaban 2011 [73]; Blanchard 2010 [74]; Brahmbhatt 2021 [75]; Figueira 2011 [76, 77]; Kim 2014 [78]; Knez 2011 [79]; Knez 2012 [80]; Liu 2017 [81]; Mahmoud 2011 [82]; NCT03659812 2018 [83]; NCT05978947 2023 [84]; Setti 2011 [85]; Setti 2012a [86, 87]; Setti 2012b [88]).

Risk of bias in included studies

For this update, we assessed the included studies for risk of bias using Cochrane's risk of bias assessment tool RoB 1 [25]. We present a brief summary of our assessments below. For further details, see [Supplementary material 2](#); [Figure 2](#); [Figure 3](#).

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies

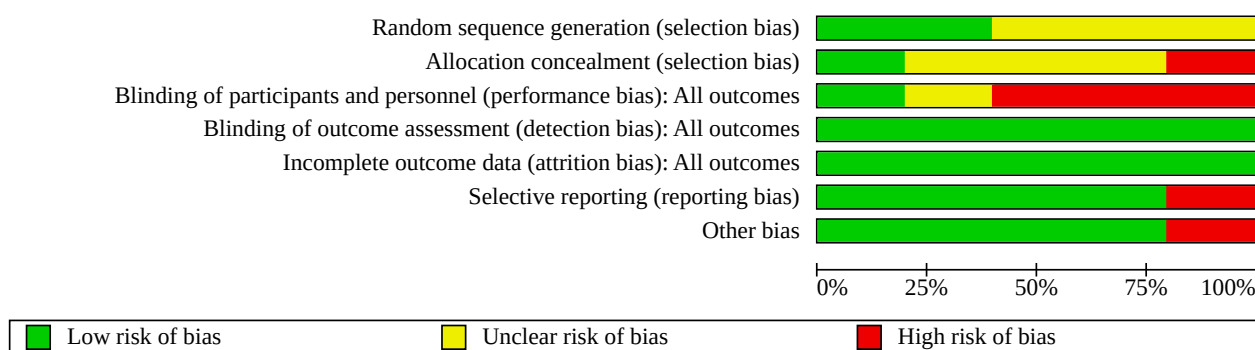


Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Esfahani 2016	?	?	+	+	+	+	+
Miller 2019	+	+	-	+	+	+	+
Parmegiani 2012a	?	?	-	+	+	+	+
WorriLOW 2013	+	-	-	+	+	-	-
Yetkinel 2019	?	?	?	+	+	+	+

Selection bias

Random sequence generation

We assessed two studies that used computer-generated randomisation at low risk of bias in relation to sequence generation (Miller 2019; Worilow 2013). Three studies were at unclear risk of bias as the method of random sequence generation was not reported (Parmegiani 2012a; Yetkinel 2019; Esfahani 2016).

Allocation concealment

We assessed one study at low risk of bias as the investigators performing randomisation had no involvement in the trial (Miller 2019), whilst information was insufficient to permit a judgement for three trials (Esfahani 2016; Parmegiani 2012a; Yetkinel 2019). One study was at high risk of bias due to our concerns that the investigators were aware of participant treatment allocation (Worilow 2013).

Performance bias

Blinding of participant and personnel

In one study, participants and personnel were blinded to the intervention each participant received (Esfahani 2016), which we assessed as low risk of bias, but three studies were at high risk of bias for this domain due to lack of clear blinding protocols (Miller 2019; Parmegiani 2012a; Worilow 2013). One study was at unclear risk of bias for this domain as blinding was not reported (Yetkinel 2019).

Detection bias

Blinding of outcome assessment

We assessed all studies at low risk of detection bias (Esfahani 2016; Miller 2019; Parmegiani 2012a; Worilow 2013; Yetkinel 2019). We judged that the lack of blinding of outcome assessment was unlikely to affect the outcomes measured.

Attrition bias

Incomplete outcome data

We assessed all studies at low risk of attrition bias (Esfahani 2016; Miller 2019; Parmegiani 2012a; Worilow 2013; Yetkinel 2019).

Four studies included data analysed as intention-to-treat with minimal to no attrition (Miller 2019; Parmegiani 2012a; Yetkinel 2019; Esfahani 2016), and one study reported clear reasons for exclusions with both groups being balanced (Worilow 2013).

Reporting bias

Selective reporting

We considered one study to be at high risk of reporting bias as data were not available for all outcome measures (Worilow 2013). We assessed the remaining studies as at low risk of reporting bias (Esfahani 2016; Miller 2019; Parmegiani 2012a; Yetkinel 2019). We found no evidence to suggest that specific outcomes were reported on the basis of statistical significance.

Other bias

One study was potentially biased as it was stopped prematurely due to financial constraints and a slower-than-expected recruitment time (Worilow 2013).

Synthesis of results

See [Summary of findings 1](#) (hyaluronic acid selected sperm (HA-ICSI) versus standard sperm selection (ICSI)); [Summary of findings 2](#) (hyaluronic acid selected sperm (HA-ICSI) versus Sperm Slow); [Summary of findings 3](#) (zeta sperm selection versus ICSI); [Summary of findings 4](#) (Microfluidic Sperm Sorting (MFSS) versus ICSI).

1. HA-ICSI versus ICSI

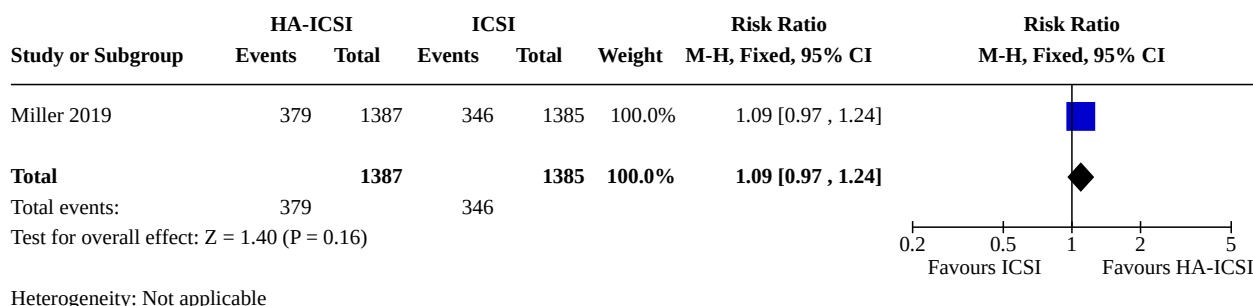
Two studies evaluated this comparison (Miller 2019; Worilow 2013).

Primary outcomes

1.1 Live birth per woman randomly assigned (effectiveness)

One study reported live birth (Miller 2019). HA-ICSI may result in no difference or a slight increase in live birth per woman randomly assigned. If ICSI leads to a 25% live-birth rate, HA-ICSI leads to a live-birth rate ranging from 24% to 31% (RR 1.09, 95% CI 0.97 to 1.24; $I^2 = 0\%$; not applicable; 1 study, 2772 participants; low-certainty evidence; [Figure 4](#); Analysis 1.1).

Figure 4. Forest plot of comparison: 1 Hyaluronic acid sperm selection (HA-ICSI) versus ICSI, outcome: 1.1 Live birth per woman randomly assigned



1.2 Miscarriage per woman randomly assigned (adverse event)

Both studies reported on miscarriage (Miller 2019; Worilow 2013). There is probably a slightly decreased rate of miscarriage with HA-

ICSI. If ICSI leads to a 7% miscarriage rate per woman, HA-ICSI leads to a rate ranging from 3% to 5% (RR 0.59, 95% CI 0.44 to 0.80; $I^2 = 0\%$;

2 studies, 3327 participants; moderate-certainty evidence; Analysis 1.2).

Secondary outcomes

1.3 Miscarriage per clinical pregnancy (adverse event)

Both studies reported on miscarriage per clinical pregnancy (Miller 2019; Worrilow 2013). There is probably a slightly decreased rate of miscarriage with HA-ICSI. If ICSI leads to an 18% miscarriage rate per clinical pregnancy, HA-ICSI leads to a rate ranging from 8% to 14% (RR 0.60, 95% CI 0.45 to 0.80; $I^2 = 0\%$; 2 studies, 1207 participants; moderate-certainty evidence; Analysis 1.3).

1.4 Clinical pregnancy per woman randomly assigned (effectiveness)

Both studies reported on clinical pregnancy (Miller 2019; Worrilow 2013). There may be little or no difference between HA-ICSI and ICSI for this outcome. If ICSI leads to a 37% clinical pregnancy rate, HA-ICSI leads to a clinical pregnancy rate ranging from 33% to 39% (RR

0.98, 95% CI 0.90 to 1.07; $I^2 = 0\%$; 2 studies, 3327 participants; low-certainty evidence; Analysis 1.4).

Foetal abnormalities (adverse event)

Neither study reported on foetal abnormalities.

2. HA-ICSI versus viscous medium containing HA (SpermSlow)

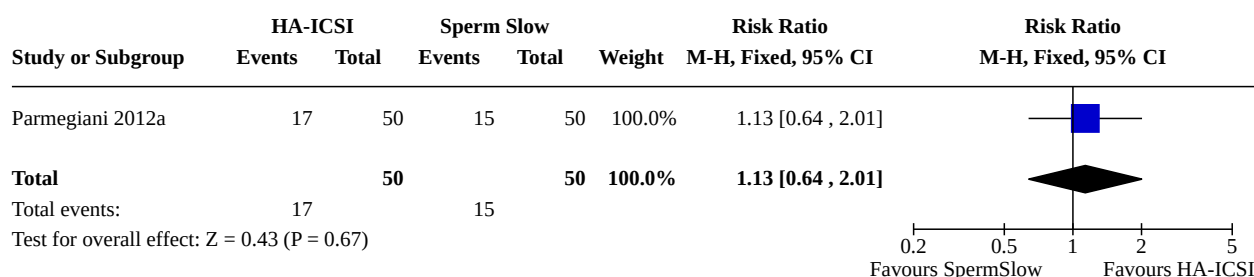
Primary outcomes

One study evaluated this comparison (Parmegiani 2012a).

2.1 Live birth per woman randomly assigned (effectiveness)

We are uncertain of the effect of the interventions on live-birth rates. If SpermSlow leads to a 30% live-birth rate, HA-ICSI leads to a live-birth rate ranging from 19% to 60% (RR 1.13, 95% CI 0.64 to 2.01; I^2 not applicable; 1 study, 100 participants; very low-certainty evidence; Figure 5; Analysis 2.1).

Figure 5. Forest plot of comparison: 2 Hyaluronic acid sperm selection (HA-ICSI) versus viscous medium containing HA (SpermSlow), outcome: 2.1 Live birth per woman randomly assigned



Heterogeneity: Not applicable

2.2 Miscarriage per woman randomly assigned (adverse event)

We are uncertain of the effect of the interventions on miscarriage rates. If SpermSlow leads to a 10% miscarriage rate per woman, HA-ICSI leads to a rate ranging from 2% to 28% (RR 0.80, 95% CI 0.23 to 2.81; I^2 not applicable; 1 study, 100 participants; very low-certainty evidence; Analysis 2.2).

Secondary outcomes

2.3 Miscarriage per clinical pregnancy (adverse event)

We are uncertain of the effect of the interventions on miscarriage per clinical pregnancy. If SpermSlow leads to a 25% miscarriage rate per clinical pregnancy, HA-ICSI leads to a rate ranging from 6% to 61% (RR 0.76, 95% CI 0.24 to 2.44; I^2 not applicable; 1 study, 41 participants; very low-certainty evidence; Analysis 2.3).

2.4 Clinical pregnancy per woman randomly assigned (effectiveness)

We are uncertain of the effect of the interventions on clinical pregnancy rates. If SpermSlow leads to a 40% clinical pregnancy

rate, HA-ICSI leads to a rate ranging from 26% to 67% (RR 1.05, 95% CI 0.66 to 1.68; I^2 not applicable; 1 study, 100 participants; very low-certainty evidence; Analysis 2.4).

Foetal abnormalities (adverse event)

Parmegiani 2012a did not report on foetal abnormalities.

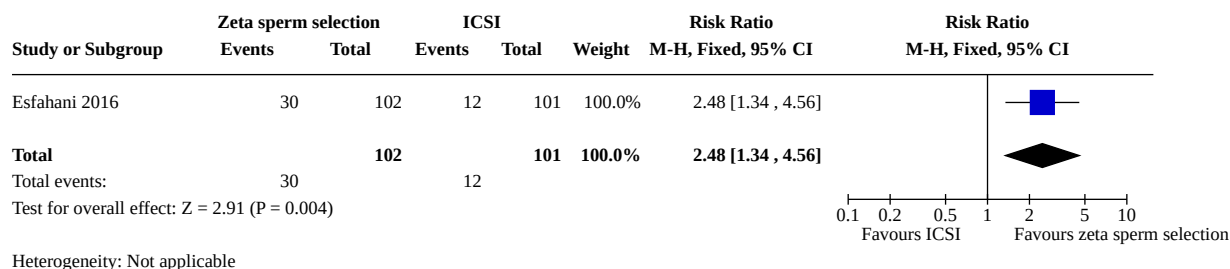
3. Zeta potential sperm selection versus ICSI

One study evaluated this comparison (Esfahani 2016).

Primary outcomes

3.1 Live birth per woman randomly assigned (effectiveness)

We are uncertain if zeta potential sperm selection improves live-birth rates. If ICSI leads to a 12% live-birth rate, zeta potential sperm selection leads to a rate ranging from 16% to 54% (RR 2.48, 95% CI 1.34 to 4.56; I^2 not applicable; 1 study, 203 participants; very low-certainty evidence; Figure 6; Analysis 3.1).

Figure 6. Forest plot of comparison: 4 Zeta sperm selection versus ICSI, outcome: 4.1 Live birth per woman randomly assigned.

3.2 Miscarriage per woman randomly assigned (adverse event)

We are uncertain of the effect of zeta potential sperm selection on miscarriage. If ICSI leads to a 4% miscarriage rate, zeta potential sperm selection leads to a rate ranging from 1% to 13% (RR 0.74, 95% CI 0.17 to 3.23; I^2 not applicable; 1 study, 203 participants; very low-certainty evidence; Analysis 3.2).

Secondary outcomes

3.3 Miscarriage per clinical pregnancy (adverse event)

We are uncertain of the effect of zeta potential sperm selection on miscarriage per clinical pregnancy. If ICSI leads to an 18% miscarriage rate per clinical pregnancy, zeta potential sperm selection leads to a rate ranging from 2% to 31% (RR 0.41, 95% CI 0.10 to 1.68; I^2 not applicable; 1 study, 62 participants; very low-certainty evidence; Analysis 3.3).

3.4 Clinical pregnancy per woman randomly assigned (effectiveness)

We are uncertain if zeta potential sperm selection improves clinical pregnancy rate. If ICSI leads to a 22% clinical pregnancy rate, zeta

potential sperm selection leads to a rate ranging from 25% to 61% (RR 1.80, 95% CI 1.16 to 2.80; I^2 not applicable; 1 study, 203 participants; very low-certainty evidence; Analysis 3.4).

Foetal abnormalities (adverse event)

Esfahani 2016 did not report on foetal abnormalities.

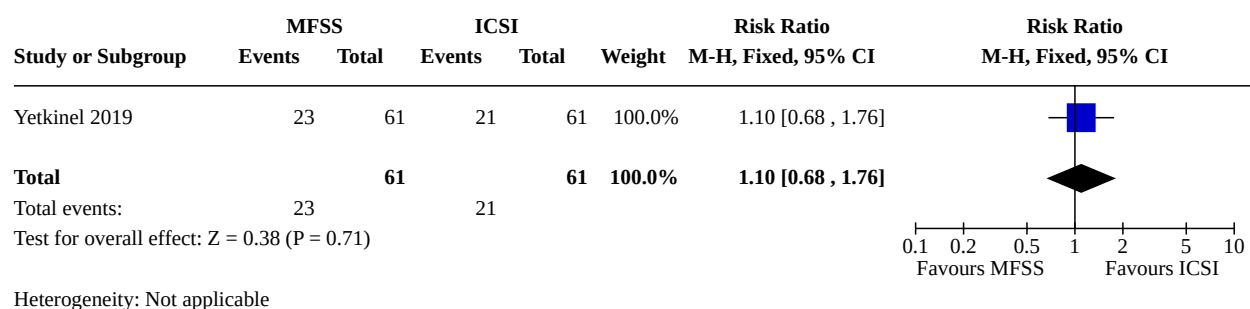
4. MFSS versus ICSI

Primary outcomes

One study evaluated this comparison (Yetkinel 2019).

4.1 Live birth per woman randomly assigned (effectiveness)

We are uncertain of the effect of MFSS on live-birth rates. If ICSI leads to a 34% live-birth rate, MFSS leads to a rate ranging from 23% to 61% (RR 1.10, 95% CI 0.68 to 1.76; I^2 not applicable; 1 study, 122 participants; very low-certainty evidence; Figure 7; Analysis 4.1).

Figure 7.

4.2 Miscarriage per woman randomly assigned (adverse event)

We are uncertain of the effect of MFSS on miscarriage. If ICSI leads to an 8% miscarriage rate, MFSS leads to a rate ranging from 3% to 30% (RR 1.20, 95% CI 0.39 to 3.72; I^2 not applicable; 1 study, 122 participants; very low-certainty evidence; Analysis 4.2).

Secondary outcomes

4.3 Miscarriage per clinical pregnancy (adverse event)

We are uncertain of the effect of MFSS on miscarriage per clinical pregnancy. If ICSI leads to a 19% miscarriage rate per clinical

pregnancy, MFSS leads to a rate ranging from 7% to 60% (RR 1.08, 95% CI 0.37 to 3.11; I^2 not applicable; 1 study, 55 participants; very low-certainty evidence; Analysis 4.3).

4.4 Clinical pregnancy per woman randomly assigned (effectiveness)

We are uncertain of the effect of Zeta sperm selection on clinical pregnancy rate. If ICSI leads to a 43% clinical pregnancy rate, MFSS leads to a rate ranging from 32% to 70% (RR 1.12, 95% CI 0.75 to 1.65; I^2 not applicable; 1 study, 122 participants; very low-certainty evidence; Analysis 4.4)

Foetal abnormalities (adverse event)

Yetkinel 2019 did not report on foetal abnormalities.

Secondary analyses

There were insufficient data to conduct subgroup analyses. We did not perform sensitivity analyses since no imputations were required, and for the main outcome of live birth, the I^2 statistic was zero, so use of the random-effects model would not have altered the effect of interventions.

Equity assessment

We did not investigate health inequity in this review as it was not part of the protocol.

Reporting biases

We did not generate funnel plots due to the small number of studies.

DISCUSSION

Summary of main results

Four of the studies included in this Cochrane review reported on live-birth rate. One study provided low-certainty evidence that there may be no difference in the live-birth rate between HA-ICSI and ICSI or a slight increase in effectiveness with HA-ICSI. We are uncertain about the results for the comparisons HA-ICSI versus SpermSlow, Zeta Sperm versus ICSI, and MFSS versus ICSI, due to the very low certainty of the evidence. A single study suggested that there may be an increased rate of live birth with the use of Zeta sperm selection compared to ICSI, but this is very low certainty evidence.

All five included studies reported on miscarriage. Based on combining the results of two studies, we found moderate-certainty evidence that HA-ICSI is probably associated with a slightly decreased risk of miscarriage when compared to standard ICSI per woman randomised and per clinical pregnancy. We are uncertain whether there is a difference in miscarriage rates for HA-ICSI versus SpermSlow, Zeta potential sperm selection versus ICSI, and MFSS versus ICSI, due to the very low certainty of the evidence.

All five included studies reported on clinical pregnancy. Based on combining the results of two studies, we found low-certainty evidence that there may be little or no difference between HA-ICSI and ICSI in clinical pregnancy rates. Due to the very low certainty of the evidence, we are uncertain whether there is a difference in clinical pregnancy rates for HA-ICSI versus SpermSlow, Zeta potential sperm selection versus ICSI, and MFSS versus ICSI.

None of the included studies reported on foetal abnormalities.

We did not identify any studies that permitted evaluation of the effect of sperm selected by sperm birefringence or MACS. None of the included studies reported a subgroup suitable for analysis. For a summary of the key results of the review, see [Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#); [Summary of findings 4](#).

Assisted reproductive technologies have drastically modified fertility potential for countless couples since their introduction. However, as these technologies develop, there appears to be

limited benefit in terms of their ability to improve outcomes. Although there is a good theoretical basis for the advanced sperm selection techniques analysed here, we did not find a technique that demonstrated a sizeable increase in the chance of successful live birth for infertility patients.

This situation may change as more evidence comes to light. The evidence does suggest that sperm selected by hyaluronic acid binding probably reduces miscarriage slightly, though it may have little or no effect on live birth or clinical pregnancy; however, we are uncertain of the effects of the other technologies evaluated in the included studies on these outcomes.

Even though we included a very large trial comparing HA-ICSI with ICSI that found a reduction in miscarriage with HA-ICSI, the trial was not powered to evaluate the outcome of miscarriage (Miller 2019). As miscarriage is a much less common outcome than live birth, the confidence interval for the absolute risk difference was smaller for miscarriage than for live birth. This might explain why the absolute improvement in live birth was not significant, even though it was similar to the absolute reduction in miscarriage. An absolute risk difference of around 2.5% has a greater effect on outcomes with a low prevalence, such as miscarriage, than on outcomes with a much higher prevalence, like live birth (Miller 2019).

Limitations of the evidence included in the review

See [Figure 2](#) and [Figure 3](#) for a summary of our assessment of the risk of bias in the included studies.

Robust conclusions on advanced sperm selection techniques to augment successful assisted reproduction technologies are not possible given the limitations of the evidence. We assessed the certainty of the evidence for the reported outcomes as moderate, low, or very low. The main limitations were poor reporting of study methods, attrition bias, potential performance bias, and imprecision due to low event rates and low participant numbers.

Regarding the outcomes of live birth and clinical pregnancy, the 95% confidence intervals could be compatible with benefit, harm, or with no effect, for all interventions except zeta potential sperm selection. Zeta potential sperm selection may improve these outcomes, yet the evidence was of very low certainty, with data derived from a single trial. We assessed the evidence for miscarriage as of moderate certainty for the comparison HA-ICSI versus ICSI, and the evidence did show a probable slight benefit for miscarriage; however, the data are limited to a single trial. For all other interventions, the 95% confidence intervals were compatible with substantial benefit or harm from the intervention, or with no effect. The available trial data for some interventions were sparse. We were unable to assess the risk of reporting bias in the body of the evidence.

It must be acknowledged that we placed a large number of studies under 'awaiting classification', which could affect the results of the review. The data in these studies could potentially impact the outcomes reported. No study reported on foetal abnormalities, which made this outcome impossible to assess. Furthermore, the changing nature of IVF and increased use of Preimplantation Genetic Testing (PGT) has resulted in an increase in frozen versus fresh cycles, and the included studies did not specify the ratio of frozen to fresh cycles (Miller 2019 included only fresh transfers; Esfahani 2016, Parmegiani 2012a, and Worilow 2013 did not clarify;

and Yetkinel 2019 used fresh transfers for the first cycle with subsequent cycles being frozen transfers). This may not reflect current practice and limits the application of results to scenarios where frozen cycles are more common [89, 90].

The objectives of this review were addressed by the included studies, which analysed relevant participants and outcomes and most of the investigations of potential relevance to clinicians. There were controlled data available to address our primary outcome measure of live birth per allocated couple, and for the secondary outcomes of clinical pregnancy and miscarriage, for some of the advanced sperm selection techniques evaluated. Data on other important clinical outcomes such as foetal abnormalities were lacking, and no suitable studies on the sperm selection technique of birefringence or MACS were found.

Limitations of the review processes

It is possible that we did not identify all available RCTs addressing the review question (including studies awaiting classification), which could bias the results of this systematic review. However, we made every effort, in line with prescribed Cochrane search strategies, to perform comprehensive searches, data collection processes, and to limit assumptions based on interventions.

Agreements and disagreements with other studies or reviews

Three other systematic reviews have addressed effects of advanced sperm selection techniques on sperm quality and ART outcomes [91, 92, 93]. A systematic review by Craciunas and colleagues evaluated HA-ICSI versus other methods for ICSI and included 11 RCTs, of which only four reported on live birth rate. The review authors concluded that there was no significant difference between sperm selected via HA-binding or polyvinylpyrrolidone (PVP) [91]. A systematic review by Said and colleagues reviewed 44 studies, including case reports and non-randomised studies, and found equivocal results [92]; they did not perform a meta-analysis to evaluate live birth rates [92]. The third systematic review, which was by Ribas-Naybou and colleagues, included 47 studies (not limited to RCTs) and reported an overall improvement in ART outcomes (30 out of 47 studies) but did not assess live birth rate specifically or perform meta-analysis [93]. IMSI, which was excluded from this review, was included in the above reviews. Further clinical trials are required before advanced sperm selection techniques can be recommended in routine practice.

AUTHORS' CONCLUSIONS

Implications for practice

The evidence for advanced sperm selection techniques during assisted reproduction is based on a small number of studies; we included only five randomised controlled trials in this updated review. Based on moderate-certainty evidence, we found that selecting sperm by hyaluronic acid binding (HA-ICSI (hyaluronic acid selected sperm-intracytoplasmic sperm injection)) probably slightly reduces miscarriage rates. Based on low-certainty evidence, we found that there may be no difference or a slight increase in live birth rate with HA-ICSI. We do not know the effects of sperm selection by zeta potential, MFSS (Microfluidic Sperm Sorting), or SpermSlow on live birth, miscarriage, and clinical pregnancy, due principally to the very low certainty of the evidence for these interventions. The evidence is therefore insufficient

to recommend routine implementation of such techniques into practice.

Further high-quality studies are required to evaluate whether any of these advanced sperm selection techniques can be recommended for use in routine practice. We are aware of 19 studies currently in progress, the results of which may allow us to reach more definitive conclusions in a future update of this review.

Equity-related implications for practice

These were not assessed in this review update.

Implications for research

Suitable randomised controlled trials (RCTs) are needed to evaluate the effects of sperm selection based on sperm birefringence and MACS (magnetic activated cell sorting) on live birth, miscarriage, and clinical pregnancy. There is a paucity of RCT data on congenital abnormalities in pregnancies utilising advanced sperm selection techniques, which represents an avenue for future RCTs to explore. It remains unclear whether certain patient subgroups, such as those with high sperm DNA fragmentation or other aetiologies of subfertility, might benefit from these advanced sperm selection techniques, which bears investigating. Trials should use intention-to-treat analysis and should report outcomes per woman randomly assigned.

Equity-related implications for research

These were not assessed in this review update.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD010461.pub4](https://doi.org/10.1002/14651858.CD010461.pub4).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Gynaecology and Fertility supported the authors in the development of this updated review.

The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Madelon van Wely, Cochrane Gynaecology and Fertility Senior Editor
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Gail Quinn, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, and supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Charalampos Siristatidis, Professor of Obstetrics, Gynecology and Reproductive Medicine, National and Kapodistrian University of Athens, Greece (clinical/content review); Sandra Ćuković (patient and public review); Lindsay Robertson, Cochrane (methods review); Jo Platt, Central Editorial Information Specialist (search review). One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

Contributions of authors

Protocol

Simon McDowell (SM) wrote the first draft of the protocol. Ben Kroon (BK) wrote the revised draft of the protocol. Emily Ford (EF) contributed methodological and statistical expertise to the protocol. Anusch Yazdani (AY) commented on all drafts of the protocol, as well as on the methods and statistics. Demian Glyjovsky (DG) assisted with revision of the protocol. Ysanne Hook (YH) provided technical expertise.

Full review (2014)

SM wrote the draft. BK provided clinical input. EF and AY supplied methodological and statistical expertise. DG commented on the revised draft. YH provided technical input.

2019 update

Sam Lepine (SL) and LS undertook abstract review and study selection. SL and SM performed data extraction. The draft was written by SL and revised by SM. SL, SM, and AY collated the response to peer review.

2026 update

Akanksha Garg (AG), Olivia Nakhala (ON), and Afsana Afroz (AA) undertook abstract review and study selection. ON, AA, SL, and Michelle Steeper (MS) performed data extraction, risk of bias assessment, and GRADE assessments of the certainty of the

What's new

evidence. The draft was written by AG and reviewed and edited by all authors.

Declarations of interest

Akanksha Garg: no conflicts of interest reported

Olivia Nakhla: no conflicts of interest reported

Michelle Steeper: no conflicts of interest reported

Afsana Afroz: no conflicts of interest reported

Sam Lepine: no conflicts of interest reported

Simon McDowell: no conflicts of interest reported

Ben Kroon: no conflicts of interest reported

Demián Glujovsky : no conflicts of interest reported

Anusch Yazdani: works as a health professional in public/private practice.

Sarah Lensen: has published and provided media commentary related to IVF add-ons.

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Internal sources

- None, Other

External sources

- None, Other

Registration and protocol

Original review (2014) DOI: 10.1002/14651858.CD010461.pub2

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library.

- Full search strategies for each database ([Supplementary material 1](#))
- Full citations of each unique report for all studies (included, ongoing, or awaiting classification, or excluded during the full-text screen ([Supplementary material 2](#); [Supplementary material 3](#); [Supplementary material 4](#); [Supplementary material 5](#)))
- Analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 6](#))

Appropriate permissions have been obtained for such use. See [Supplementary material 7](#) for the complete data package. Analyses and data management were conducted within Cochrane's authoring tool, RevMan [27], using the inbuilt computation methods.

Date	Event	Description
2 June 2026	New search has been performed	Search updated. We included one new study (Yetkinel 2019). A new comparison (Microfluidic Sperm Sorting (MFSS) versus ICSI) has been added to the review. Due to our concerns about the trustworthiness of their data, we removed four studies that were included in the previous version of the review to 'awaiting classification' (Majumdar 2013; Romany 2014; Troya 2015; Ziarati 2018).
2 June 2026	New citation required and conclusions have changed	The removal of four studies has led to changes in our conclusions. In the HA-ICSI vs ICSI comparison, the certainty of evidence for miscarriage has increased from low to moderate certainty evidence. HA-ICSI probably reduces miscarriage rates and makes no difference to or results in a slight increase in live birth rates. The evidence for other techniques is limited.

History

Protocol first published: Issue 3, 2013

Review first published: Issue 10, 2014

Date	Event	Description
21 March 2019	New search has been performed	Review updated. We included 6 new studies (Esfahani 2016; Majumdar 2013; Miller 2019; Romany 2014; Troya 2015; Ziarati 2018).
21 March 2019	New citation required and conclusions have changed	The addition of 6 new studies has led to changes in conclusions.

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ADDITIONAL TABLES

Advanced sperm selection techniques for assisted reproduction (Review)

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Table 1. Overview of included studies

Study ID / Country	Study design	Population	Intervention	Comparator	Main outcomes	Other important outcomes
Esfahani 2016 Iran	RCT	- Women: below 40 years with adequate number of follicles - Men: at least 1 abnormal semen parameter of their partner	Zeta sperm selection + DGC + ICSI (n = 101)	DGC+ ICSI (n = 102)	- Live birth - Clinical pregnancy - Miscarriage	- Implantation rate - Fertilisation rate - Top quality embryos
Miller 2019 UK	RCT	- Women: 18 to 43 years, BMI 19 to 35, FSH 3 to 20, AMH at least 1.5 - Men: 18 to 55 years, no vasovasostomy, no cancer, 3 days of abstinence	HA-ICSI (n = 1387)	ICSI (n = 1385)	- Live birth - Clinical pregnancy - Miscarriage	Fertilisation rate
Parmegiani 2012 Italy	RCT	- Women: Infertile; aged ≤ 41 years; undergoing ICSI treatment - Men: total sperm number ≥ 1 million; sperm motility $\geq 5\%$	HA-ICSI (n = 50)	SpermSlow+ ICSI (n = 50)	- Live birth - Clinical pregnancy - Miscarriage	- Implantation rate - Fertilisation rate - Good quality embryos
Worriolow 2013 USA	RCT	Couples receiving ICSI	HA-ICSI (n = 276)	ICSI (n = 279)	- Clinical pregnancy - Miscarriage	Implantation rate
Yetkinel 2019 Israel	RCT	- Couples with unexplained infertility - Women: between 20 and 37 years of age; normal tubal patency, normal uterine cavity, normal ovulation - Men: between 20 to 40 years of age and normal semen analysis by WHO criteria	Microfluidic system + ICSI (n = 61)	Swim up + ICSI (n = 61)	- Live birth - Clinical pregnancy - Miscarriage	Fertilisation rate

AMH: anti-mullerian hormone; **BMI:** body mass index; **DGC:** density gradient centrifugation; **FSH:** follicle stimulating hormone; **ICSI:** intracytoplasmic sperm injection; **HA-ICSI:** hyaluronic acid selected sperm-intracytoplasmic sperm injection; **RCT:** randomised controlled trial; **WHO:** World Health Organization